A Guided Reference to Bogolepov, Lipatov & Sokolov (1990): *Structure of Chemically Nonequilibrium Flows with a Sudden Change in the Temperature and the Catalytic Properties of the Surface*

Explanatory material and a full equation, numerical-results, and numerical-implementation compendium prepared as a reference companion to Bogolepov, V. V., Lipatov, I. I. & Sokolov, L. A. (1990), *J. Appl. Mech. Tech. Phys.* **31**(3), 367–377 (translated from *Zh. Prikl. Mekh. Tekh. Fiz.*, No. 3, pp. 30–41, 1990).

Abstract

Bogolepov, Lipatov & Sokolov (1990) pack a four-region asymptotic structure, two self-similar eigenvalue problems, and a set of skin-friction and heat-flux predictions into eleven pages; this companion unpacks that paper in three parts. **Part I** is explanatory. It is written for a reader new to hypersonic aerothermodynamics, chemical nonequilibrium flow, or matched asymptotic expansions, and it does not follow the source paper's wording. **Part II** is a faithful compendium of every numbered equation and every reported numerical result, filed under the paper's own section and equation numbers so that it can serve directly as a working reference. **Part III** is where this document goes beyond its source: it records a MATLAB implementation of the downstream self-similar eigenvalue problem (eq. 2.28), the corrections to the governing equations that the implementation exposed, and the current validation status. One thing is deliberately left out. The paper's own prose (its literature review, its sentence-level argument, and its figures as drawn) is copyrighted text, and Part I is written precisely so that it is not needed — everything that prose did to orient the reader is done here again, in a clearer and self-contained way.

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Part I
Understanding the Problem: A Guided Introduction

1 The physical picture in one paragraph

Picture a flat plate flying through the air at several times the speed of sound. Right at the plate's surface, a thin sluggish layer of gas — the boundary layer — clings to the plate because of viscosity; outside that layer the gas screams past essentially inviscidly. Now suppose the plate is made of two different materials glued together end to end: the front half is one material, and at some point along the plate, the surface abruptly becomes a different material with a different temperature and a different chemical behaviour. This paper asks a very specific, very local question: what does the flow do in the immediate vicinity of that seam? Not far upstream, not far downstream — but in the small stretch of plate right around the jump, where the boundary layer has to somehow reconcile what it was doing before the jump with what the new surface demands of it afterward.

The answer is a cascade of nested "zoom levels". Each is a region of the flow at a progressively smaller length scale, each governed by a simplified version of the full equations that is valid at that particular scale. Building and matching these nested regions is the technique called *matched asymptotic expansions*, and it is the paper's central methodological tool. The rest of this introduction builds up the vocabulary needed to follow that construction.

2 Why the gas's chemistry matters here at all

At the speeds this paper is concerned with (uniform supersonic flow, with the boundary layer itself reaching temperatures high enough to matter chemically), the air is modelled as a simplified two-species gas: a **binary mixture of atoms and the diatomic molecules made of those same atoms** — think of it as a stand-in for, say, atomic oxygen (O) mixed with molecular oxygen (O_2), without worrying about nitrogen or the dozen other species real air actually contains. It is a standard, well-worn simplification in hypersonic aerothermodynamics. It keeps just enough chemistry to capture the essential physics (dissociation and recombination) without the bookkeeping of a full multi-species reacting-gas model, and for a paper whose real subject is asymptotic structure rather than chemical kinetics, that trade is the right one.

2.1 Equilibrium, nonequilibrium, and "frozen" flow

A reacting mixture can sit in one of three regimes. Which one applies decides which equations are worth writing down at all:

- **Chemical equilibrium.** The chemical reaction (here, molecules splitting into atoms, and atoms recombining into molecules) happens so *fast* compared to how quickly the gas is moving and changing conditions that, at every point, the mixture composition is always exactly what it "should" be for the local temperature and pressure — as if the flow could be paused at any point and the chemistry would already have settled into its textbook equilibrium ratio for those conditions.

- **Nonequilibrium flow.** The reaction rate and the flow's own rate of change are comparable, so the local composition *lags behind* what equilibrium would dictate. This requires actually solving a differential equation for how the atom concentration evolves along each fluid path, with a genuine chemical source term in it (a term describing how fast atoms are being created or destroyed by the reaction).

- **"Frozen" flow.** The reaction is so *slow* compared to the flow that, to the accuracy needed, the composition simply does not change at all as a fluid particle moves along — it is "frozen" at whatever composition it had, carried along unchanged. Chemistry still matters (a frozen mixture of atoms and molecules still behaves differently, thermodynamically, from a mixture that has recombined), but the *rate* of reaction drops out of the equations entirely in the bulk of the flow.

This paper works in the frozen-flow limit. Rather than simply assuming it, the paper derives *why* it is the right limit for the small region it studies, from the scaling estimates of its Section 1 (collected in Part II, §5 below). The short version: right at the jump in surface properties, the flow gradients become so steep that everything else in the equations becomes far bigger than the chemical reaction-rate term, which becomes negligible by comparison — even though, ahead of and behind this tiny region, the gas may not be frozen at all in the ordinary boundary layer.

2.2 What "catalytic surface" and "catalycity" mean here

Even in frozen flow, the composition can still change at one special place: the wall. If atoms happen to touch a solid surface that is chemically active toward them, they can recombine into molecules right there, at the surface, releasing energy (heat) as they do — this is exactly the same physical idea as a catalytic converter in a car, or a catalytic recombination coating on a spacecraft heat shield. A surface's *catalycity* (or catalytic efficiency) is simply a number describing how readily atoms that reach the surface recombine there, on a scale from *non-catalytic* (atoms bounce off unchanged, essentially never recombining) to *ideally catalytic* (every atom that touches the wall instantly recombines).

This matters because recombination at the wall releases a great deal of heat. It is one of the primary reasons hypersonic vehicle surfaces get so hot, and a major reason why heat-shield material science cares about controlling a surface's catalycity. In the paper's mathematics, catalycity shows up as a specific boundary condition at the wall (Part II, eq. 2.14 below) that relates the local atom concentration and its gradient near the wall to a catalytic-rate coefficient k — and the paper's whole subject is what happens when that coefficient, and the wall temperature, both jump discontinuously at one point along the plate.

2.3 Why a density change near the wall matters mechanically

One subtlety is easy to miss, and the paper is right to flag it: when the surface catalycity changes (say, from non-catalytic to catalytic), atoms near the wall recombine into denser molecules, so the gas density right next to the wall jumps up. A denser gas, at the same mass flow rate, occupies less volume — so the streamlines just above the wall get pulled down, toward the surface, as if the wall had suddenly grown a shallow dimple or cavity that the flow has to dip into. (The reverse happens if the temperature jumps up instead: hotter gas is less dense, so the streamlines are pushed away from the surface, as if the wall had suddenly grown a shallow bump.) This "effective geometry" — a virtual dimple or bump that the outer flow does not know is virtual — is precisely what generates a pressure disturbance that propagates back into the outer supersonic stream, and it is the mechanism that ties this problem to the much more general subject of flow over small surface irregularities (the same underlying mathematics, in fact, that governs flow over a literal small bump or dent on a wall).

3 Why multiple nested regions are needed: the logic of matched asymptotic expansions

Suppose one tried to write down a single equation valid everywhere near the jump. Trouble would follow immediately: right at the wall, viscosity and heat conduction and mass diffusion all matter enormously; a little farther out, in the bulk of the boundary layer, they barely matter at all; and farther out still, outside the boundary layer altogether, the flow is essentially frictionless supersonic flow. No single simplified equation is accurate at all of these places simultaneously.

The method of *matched asymptotic expansions* solves this by admitting defeat on that goal and doing something cleverer instead: carve the flow into several distinct *regions*, each at its own characteristic length scale, in each of which a simpler, scale-appropriate set of equations can be written down; solve each region's simplified problem; and then insist that neighbouring regions' solutions agree with each other in the zone where their ranges of validity overlap (this is the "matching" step). Done correctly, the patchwork of regional solutions reconstructs the full behaviour of the

flow everywhere, even though no single region's equations would have been valid everywhere on their own.

The parameter that controls how thin the boundary layer is, and hence how many nested regions are needed and how they scale, is

$$\varepsilon = Re_0^{-1/2},$$

built from the (large) Reynolds number Re_0 based on distance from the leading edge — so $\varepsilon \to 0$ is simply "the boundary layer becomes vanishingly thin compared with the distance travelled," which is the everyday hypersonic/high-speed limit. Every region in this paper is labelled by how its own thickness and length scale with powers of ε.

3.1 The estimate that sets the whole problem's scale

Before building any region in detail, the paper first asks a simple question: near the jump, if the wall layer's thickness changes by an amount Δy over a stretch of plate of length Δx, how are Δy and Δx related? Balancing how fast momentum is carried along the layer (convection) against how fast viscosity smooths out the resulting velocity differences (diffusion) gives the estimate

$$\Delta y \sim O(\varepsilon\, \Delta x^{1/3}) \leq O(\varepsilon).$$

A change in the wall layer's thickness pushes the boundary layer's outer edge in or out by a proportional amount, and that displacement, felt by the outer supersonic flow, induces a pressure disturbance $\Delta p \sim O(\Delta y/\Delta x)$. If, as is generally the case here, that pressure disturbance is strong enough to produce a genuinely nonlinear velocity response in the wall layer (rather than a negligible tweak), a self-consistent balance of all these estimates together pins down a single, distinguished size for the disturbed region:

$$\Delta x \sim O(\varepsilon^{3/4}), \qquad \Delta y \sim O(\varepsilon^{5/4}), \qquad \Delta p \sim O(\varepsilon^{1/2}).$$

Every region constructed in the rest of the paper is built around this one estimate — it is the mathematical fingerprint of the physical statement "the disturbance from the jump is confined to a small, specific neighbourhood, and within that neighbourhood the induced pressure feedback genuinely matters and cannot be ignored." The underlying idea, that a disturbance's induced pressure feeds back nonlinearly on the very sublayer that produced it, is the one behind *free-interaction* or *triple-deck* theory, best known for what it says about the structure of boundary-layer separation. Here it is put to a less familiar use: the "obstacle" is not a corner or a hump but a change in surface chemistry and temperature.

3.2 The four regions, in plain language

With that scale established, the paper builds four nested regions (numbered I through IV in the paper; this document keeps that numbering):

Region I — the outer supersonic stream. Far enough from the wall (though still close to the plate on the scale of the whole vehicle), the gas behaves as an ordinary frictionless, chemically frozen supersonic flow, only slightly disturbed by whatever is happening at the wall far below it. Mathematically, this becomes the classical linearized supersonic-flow (wave) equation — the same equation used for any small disturbance to a uniform supersonic stream, with no chemistry or viscosity in it at all.

Region II — the undisturbed part of the boundary layer. This is the bulk of the boundary layer, at its normal (pre-jump) thickness, just perturbed slightly by whatever is going on right at the wall beneath it. Its role is almost bookkeeping: it passively transmits the wall disturbance's effect out to region I, and its equations can be solved in closed form once the shape of the undisturbed boundary-layer profile it started with is known.

Region III — the reacting, heat-conducting, diffusing wall layer. This is where the real physics of the problem lives: a thin sublayer, thinner than the ordinary boundary layer, where viscosity, heat conduction, and mass diffusion of the atomic species all matter simultaneously, and where the wall's jump in temperature and catalycity is directly felt. It spans the whole extent of the disturbed region (both upstream and downstream of the jump point) and is where the induced-pressure feedback from region I actually enters the sublayer's own momentum balance.

Region IV — the immediate neighbourhood of the jump itself. Region III's own equations, it turns out, are not smooth enough right at the jump point to be solved outright — approaching the jump from downstream, the solution needs to be examined on a yet-smaller length scale, tucked entirely inside region III, in order to properly connect the flow just ahead of the jump to the flow just behind it.

Put loosely, region I is the "weather" (the ambient supersonic flow, barely aware anything is happening); region II is the "commute," passing information between the wall and the weather without much drama of its own; region III is where the jump's effects actually live and evolve, spanning the whole disturbed neighbourhood; and region IV is a magnifying glass held over the single most delicate point inside region III, the jump itself.

3.3 The recurring idea of a "compensation regime"

One result deserves flagging in advance, because in the equations alone it is easy to read straight past it. Right at region IV (the jump point), the paper shows that the geometry disturbance created there is exactly cancelled, to leading order, by an equal and opposite change in the shape of the flow immediately upstream of it, inside region III — so that, remarkably, no pressure disturbance is transmitted out to the outer supersonic stream from this innermost region at all. This is called a *compensation regime*: two effects (here, the new local flow pattern right at the jump, and the adjustment of the flow just ahead of it) offset each other so precisely that their combined footprint on the outer flow vanishes at leading order. The motif recurs throughout this style of asymptotic theory. Here it earns its keep by letting the mathematics right at the jump stay tractable (technically, it keeps the problem *parabolic* — solvable by marching forward in x from initial data — rather than the harder, globally-coupled *elliptic* problem that would otherwise arise).

4 What the paper ultimately computes

Once the four regions are built and matched, everything reduces to one core numerical problem (Part II, eq. 2.25): a set of coupled ordinary/partial differential equations for the local velocity, temperature, and atom-concentration profiles near the jump, together with the induced pressure, to be solved by marching from far upstream, through the jump, out to far downstream. Two further self-contained sub-problems (eqs. 2.27–2.28 and 3.5–3.6) describe, respectively, how the solution behaves very far downstream of the jump and immediately at the jump point itself — each of these smaller problems has its own extra unknown constant (an *eigenvalue*, in the jargon), which is not

given in advance but instead falls out of demanding that the sub-problem have a physically sensible (bounded, well-behaved) solution at all.

The physical payoff is a set of concrete predictions: how the induced pressure, the wall shear stress (skin friction), and the wall's normal temperature and atom-concentration gradients (which set the heat flux and mass-transfer rate) all vary through the neighbourhood of the jump, for three representative combinations of "how much hotter" and "how much more catalytic" the downstream surface is. Part II §6 below reports these results in full.

Part II
Full Equation and Numerical-Results Compendium

Equation numbers below match the source paper exactly (Sections 1–4 there correspond to Sections 5–8 here). All quantities are nondimensionalized as in the paper: lengths by the distance s from the leading edge to the jump point, velocities by the freestream velocity u_0, pressure and shear stress by $2\rho_0 u_0^2$ (twice the freestream dynamic head), enthalpy by u_0^2, temperature by mu_0^2/R (m the molecular weight of the molecular species, R the universal gas constant), heat flux by $\rho_0 u_0^3$, the catalytic coefficient by u_0, specific heats by R/m, and all other flow quantities by their freestream values. The Reynolds number is $Re_0 = \rho_0 u_0 s/\mu_0 = \varepsilon^{-2} \to \infty$.

5 Section 1: Scaling estimates

Boundary-layer thickness variation as a function of its streamwise extent $\Delta x \le O(1)$:

$$\Delta y \sim O(\varepsilon\, \Delta x^{1/3}) \le \delta. \tag{1.1}$$

Distinguished (self-consistent, nonlinear-interaction) scaling of the disturbed region:

$$\Delta x \sim O(\varepsilon^{3/4}), \qquad \Delta y \sim O(\varepsilon^{5/4}), \qquad \Delta p \sim O(\varepsilon^{1/2}). \tag{1.2}$$

6 Section 2: The four-region asymptotic structure

Region 1 (outer uniform supersonic flow)

Coordinates and expansions:

$$x = 1 + \varepsilon^{3/4} x_1, \qquad y = \varepsilon^{3/4} y_1, \qquad u = 1 + \varepsilon^{1/2} u_1 + \cdots, \qquad v = \varepsilon^{1/2} v_1 + \cdots, \tag{2.1}$$

$$p = p_0 + \varepsilon^{1/2} p_1 + \cdots, \qquad \rho = 1 + \varepsilon^{1/2} \rho_1 + \cdots, \qquad h = h_0 + \varepsilon^{1/2} h_1 + \cdots. \tag{1}$$

Linearized Euler equations (frozen chemical composition):

$$\frac{\partial \rho_1}{\partial x_1} + \frac{\partial u_1}{\partial x_1} + \frac{\partial v_1}{\partial y_1} = 0, \qquad \frac{\partial u_1}{\partial x_1} + \frac{\partial p_1}{\partial x_1} = 0, \qquad \frac{\partial h_1}{\partial x_1} - \frac{\partial p_1}{\partial x_1} = 0. \tag{2.2}$$

Frozen-composition ("frozen") Mach number and enthalpy-increment relations:

$$M^2 = \frac{1 - \partial h/\partial p}{\partial h/\partial \rho}, \qquad h_1 = p_1 \frac{\partial h}{\partial p} + \rho_1 \frac{\partial h}{\partial \rho}. \tag{2.3}$$

d'Alembert reduction to the wave equation; outer relation as $y_1 \to 0$:

$$\sqrt{M^2-1}\, p_1(x_1,0) = v_1(x_1,0). \tag{2.4}$$

Region 2 (main, undisturbed part of the boundary layer)

Coordinates and expansions ($u_{20}, \rho_{20}, T_{20}, c_{20}$ the undisturbed boundary-layer profiles ahead of the jump):

$$x = 1+\varepsilon^{3/4}x_2, \qquad y = \varepsilon y_2, \qquad u = u_{20}(y_2)+\varepsilon^{1/4}u_{21}+\cdots, \qquad v = \varepsilon^{1/2}v_{21}+\cdots, \tag{2.5}$$

$$\rho = \rho_{20}(y_2)+\varepsilon^{1/4}\rho_{21}+\cdots, \qquad p = p_0+\varepsilon^{1/2}p_2+\cdots,$$
$$T = T_{20}(y_2)+\varepsilon^{1/4}T_{21}+\cdots, \qquad c = c_{20}(y_2)+\varepsilon^{1/4}c_{21}+\cdots. \tag{2}$$

Resulting linear system, from the Navier–Stokes and mass-conservation equations:

$$\rho_{20}\frac{\partial u_{21}}{\partial x_2}+u_{20}\frac{\partial \rho_{21}}{\partial x_2}+\rho_{20}\frac{\partial v_{21}}{\partial y_2}+v_{21}\frac{\mathrm{d}\rho_{20}}{\mathrm{d}y_2}=0, \qquad \frac{\partial p_2}{\partial y_2}=0, \tag{2.6a}$$

$$u_{20}\frac{\partial u_{21}}{\partial x_2}+v_{21}\frac{\mathrm{d}u_{20}}{\mathrm{d}y_2}=0, \qquad u_{20}\frac{\partial T_{21}}{\partial x_2}+v_{21}\frac{\mathrm{d}T_{20}}{\mathrm{d}y_2}=0, \qquad u_{20}\frac{\partial c_{21}}{\partial x_2}+v_{21}\frac{\mathrm{d}c_{20}}{\mathrm{d}y_2}=0, \tag{3}$$

with partial-integral solution (arbitrary function $D = D(x_2)$):

$$u_{21} = D\frac{\mathrm{d}u_{20}}{\mathrm{d}y_2}, \quad v_{21} = -u_{20}\frac{\mathrm{d}D}{\mathrm{d}x_2}, \quad \rho_{21} = D\frac{\mathrm{d}\rho_{20}}{\mathrm{d}y_2},$$
$$T_{21} = D\frac{\mathrm{d}T_{20}}{\mathrm{d}y_2}, \quad c_{21} = D\frac{\mathrm{d}c_{20}}{\mathrm{d}y_2}, \quad p_2 = p_2(x_2). \tag{2.6}$$

Region 3 (viscous, heat-conducting, diffusive wall layer)

Coordinates and expansions:

$$x = 1+\varepsilon^{3/4}x_3, \qquad y = \varepsilon^{5/4}y_3, \qquad u = \varepsilon^{1/4}u_3+\cdots, \qquad v = \varepsilon^{3/4}v_3+\cdots, \tag{2.7}$$

$$\rho = \rho_3+\cdots, \qquad p = p_0+\varepsilon^{1/2}p_3+\cdots, \qquad T = T_{30}+\varepsilon^{1/4}T_{31}+\cdots, \qquad c = c_{30}+\varepsilon^{1/4}c_{31}+\cdots. \tag{4}$$

Compressible boundary-layer system with species diffusion (Pr the Prandtl number, Sc the Schmidt number, c_p, c_{p1}, c_{p2} the specific heats of the mixture, atomic, and molecular components):

$$\frac{\partial(\rho_3 u_3)}{\partial x_3}+\frac{\partial(\rho_3 v_3)}{\partial y_3}=0, \qquad \frac{\partial p_3}{\partial y_3}=0, \qquad p_3 = \rho_3 T_{30}(1+c_{30}), \tag{2.8a}$$

$$\rho_3 u_3\frac{\partial u_3}{\partial x_3}+\rho_3 v_3\frac{\partial u_3}{\partial y_3}+\frac{\mathrm{d}p_3}{\mathrm{d}x_3}=\frac{\partial}{\partial y_3}\left(\mu\frac{\partial u_3}{\partial y_3}\right), \tag{5}$$

$$\rho_3 u_3\frac{\partial c_{30}}{\partial x_3}+\rho_3 v_3\frac{\partial c_{30}}{\partial y_3}=\frac{\partial}{\partial y_3}\left(\frac{\mu}{Sc}\frac{\partial c_{30}}{\partial y_3}\right), \tag{6}$$

$$\rho_3 c_p u_3 \frac{\partial T_{30}}{\partial x_3} + \rho_3 c_p v_3 \frac{\partial T_{30}}{\partial y_3} + \frac{\mathrm{d}p_3}{\mathrm{d}x_3} = \frac{\partial}{\partial y_3}\left(\frac{\mu}{Pr}\frac{\partial T_{30}}{\partial y_3}\right) + (c_{p1} - c_{p2})\frac{\mu}{Pr}\frac{\partial T_{30}}{\partial y_3}\frac{\partial c_{30}}{\partial y_3}. \tag{2.8}$$

(In this region, barodiffusion and streamwise mass diffusion are both negligible, by the estimates of §1.)

Outer (far-field) matching conditions, combining regions 1–3 ($A = \mathrm{d}u_{20}/\mathrm{d}y_2$ at $y_2 = 0$):

$$u_3 \to A(y_3 + D), \qquad T_{30} \to T_{20}(0), \qquad c_{30} \to c_{20}(0), \qquad y_3 \to \infty. \tag{2.9}$$

Pressure-matching (interaction) condition across regions 1–3:

$$p_3(x_3) = p_2(x_2) = p_1(x_1, 0) = \frac{v_{21}(x_2, \infty)}{\sqrt{M^2 - 1}} = -\frac{\mathrm{d}D/\mathrm{d}x_2}{\sqrt{M^2 - 1}}. \tag{2.10}$$

Upstream (initial) matching conditions:

$$u_3 \to Ay_3, \qquad T_{30} \to T_{20}(0), \qquad c_{30} \to c_{20}(0), \qquad p_3, D \to 0, \qquad x_3 \to -\infty. \tag{2.11}$$

No-slip and impermeability at the wall:

$$u_3 = v_3 = 0, \qquad y_3 = 0. \tag{2.12}$$

Temperature jump condition at the wall ($\alpha > -1$ the size of the temperature jump):

$$T_{30} = T_{20}(0) \ (x_3 < 0), \qquad T_{30} = (1 + \alpha)T_{20}(0) \ (x_3 \geq 0, \ y_3 = 0). \tag{2.13}$$

Catalytic-surface boundary condition for the atomic mass concentration ($k = \varepsilon^\beta K$, $K \sim O(1)$, the catalycity coefficient):

$$\frac{\partial c}{\partial y} = \frac{Sc\, k \rho c}{\varepsilon^2}, \qquad y = 0. \tag{2.14}$$

Non-ideally-catalytic upstream condition ($\beta = 1$, valid provided $c_{20}(0) > 0$ ahead of the jump):

$$\frac{\partial c_{30}}{\partial y_3} = 0, \qquad x_3 < 0, \ y_3 = 0. \tag{2.15}$$

Resulting exact upstream solution ($x_3 < 0$):

$$T_{30}(x_3, y_3) = T_{20}(0), \qquad c_{30}(x_3, y_3) = c_{20}(0). \tag{2.16}$$

Downstream ($x_3 \geq 0$) catalytic boundary condition, valid for $\beta \leq 3/4$:

$$\frac{\partial c_{30}}{\partial y_3} = Sc\, K \rho_3 c_{30} \quad \text{or} \quad c_{30} = 0, \qquad x_3 \geq 0, \ y_3 = 0. \tag{2.17}$$

Next-order (first-correction) linear system for T_{31}, c_{31}:

$$\rho_3 u_3 \frac{\partial c_{31}}{\partial x_3} + \rho_3 v_3 \frac{\partial c_{31}}{\partial y_3} = \frac{\partial}{\partial y_3}\left(\frac{\mu}{Sc}\frac{\partial c_{31}}{\partial y_3}\right),$$

$$\rho_3 c_p u_3 \frac{\partial T_{31}}{\partial x_3} + \rho_3 c_p v_3 \frac{\partial T_{31}}{\partial y_3} = \frac{\partial}{\partial y_3}\left(\frac{\mu}{Pr}\frac{\partial T_{31}}{\partial y_3}\right). \tag{2.18}$$

Outer matching conditions for T_{31}, c_{31} ($B = \mathrm{d}T_{20}/\mathrm{d}y_2$, $C = \mathrm{d}c_{20}/\mathrm{d}y_2$ at $y_2 = 0$):

$$T_{31} \to B(y_3 + D), \qquad c_{31} \to C(y_3 + D), \qquad y_3 \to \infty. \tag{2.19}$$

Upstream (initial) matching conditions:

$$T_{31} \to By_3, \qquad c_{31} \to Cy_3, \qquad x_3 \to -\infty. \tag{2.20}$$

Wall conditions for T_{31}, c_{31} ($x_3 < 0$):

$$T_{31} = 0, \qquad \frac{\partial c_{31}}{\partial y_3} = C, \qquad y_3 = 0. \tag{2.21}$$

Viscous shear stress and heat flux in region 3 (Le the Lewis–Semenov number, h_0 the dimensionless heat of atom formation):

$$\tau = \varepsilon\mu \frac{\partial u_3}{\partial y_3} + \cdots, \qquad -q = \varepsilon^{3/4} q_1 + \varepsilon q_2 + \cdots, \tag{2.22}$$

$$q_1 = \frac{\mu}{Pr}\left[c_p \frac{\partial T_{30}}{\partial y_3} + Le\, \frac{\partial c_{30}}{\partial y_3}\left(\int_0^{T_{30}} (c_{p1} - c_{p2})\,\mathrm{d}T + h_0\right)\right], \tag{7}$$

$$q_2 = \frac{\mu}{Pr}\left[c_p \frac{\partial T_{31}}{\partial y_3} + Le\, \frac{\partial c_{31}}{\partial y_3}\left(\int_0^{T_{30}} (c_{p1} - c_{p2})\,\mathrm{d}T + h_0\right)\right]. \tag{8}$$

Reduction to standard (numerical) form

Change of variables:

$$x = \frac{x_3}{x^*}, \quad \eta = \frac{1}{\eta^*}\int_0^{y_3} \rho_3\,\mathrm{d}y_3, \quad \psi = \frac{\psi_3}{\psi^*}, \quad \frac{\partial \psi_3}{\partial y_3} = \rho_3 u_3, \quad \frac{\partial \psi_3}{\partial x_3} = -\rho_3 v_3, \tag{2.23a}$$

$$p = \frac{p_3}{p^*}, \quad c_1 = c_{30}, \quad c_2 = \frac{c_{31}\rho^*}{C\eta^*}, \quad d = \frac{D\rho^*}{\eta^*}, \quad T_1 = \frac{T_{30}}{T_{20}(0)[1 + c_{20}(0)]}, \quad T_2 = \frac{T_{31}\rho^*}{B\eta^*}, \tag{9}$$

$$\rho^* = \frac{p_0}{T_{20}(0)[1 + c_{20}(0)]}, \quad \eta^* = (\rho^*)^{1/4}(M^2 - 1)^{-1/8} A^{-3/4}, \quad x^* = (\rho^*)^{-1/4}(M^2 - 1)^{-3/8} A^{-5/4}, \tag{10}$$

$$\psi^* = (\rho^*)^{-1/2}(M^2 - 1)^{-1/4} A^{-1/2}, \qquad p^* = (\rho^*)^{-1/2}(M^2 - 1)^{-1/4} A^{1/2}. \tag{2.23}$$

Simplifying assumptions:

$$\rho_3\mu = 1, \qquad Pr, Sc, c_p = \mathrm{const}, \qquad \int_0^{T_{30}} (c_{p1} - c_{p2})\,\mathrm{d}T \ll h_0. \tag{2.24}$$

Standard-form boundary value problem (the central governing system for the whole disturbed neighbourhood):

$$\psi''' = -\ddot{d}\,(1 + c_1)T_1 + \psi\dot{\psi}' - \dot{\psi}\psi'', \qquad \frac{c_i''}{Sc} = \psi\dot{c}_i - \dot{\psi}c_i', \qquad \frac{T_i''}{Pr} = \psi\dot{T}_i - \dot{\psi}T_i', \qquad i = 1, 2, \tag{2.25a}$$

$$\psi(x,0)=0, \quad \psi'(x,0)=0, \quad c_1(x,0)=F, \quad c_2(x,0)=1, \tag{11}$$

$$T_1(x,0)=\frac{c_1(x,0)}{1+c_1(x,0)}, \quad T_2(x,0)=0 \ (x<0); \qquad T_1(x,0)=\frac{1+\alpha}{1+c_{20}(0)}, \quad T_2(x,0)=0 \ (x\geq 0), \tag{12}$$

$$\psi'(x,\infty)\to\int_0^\infty T_1(1+c_1)\,\mathrm{d}\eta+d, \qquad c_1(x,\infty)\to c_{20}(0), \qquad T_1(x,\infty)\to\frac{1}{1+c_{20}(0)}, \tag{13}$$

$$T_2(-\infty,\eta)\to\eta, \quad d(-\infty)\to 0, \quad c_1(-\infty,\eta)\to c_{20}(0), \quad c_2(x,\infty)\to 1, \quad T_2(x,\infty)\to 1, \tag{14}$$

$$T_1(x,\eta)=\frac{1}{1+c_{20}(0)} \ (x<0), \tag{2.25}$$

where $(')=\partial/\partial\eta$, $(\dot{})=\partial/\partial x$, and

$$F=\frac{Sc\,K\,p_0^{5/4}}{(M^2-1)^{1/8}A^{3/4}\,T_{20}(0)(1+\alpha)[1+c_{20}(0)]^{1/4}} \tag{15}$$

is the local Damköhler number (the ratio of the reaction rate to the flow rate right at the jump; $F\to 0$ recovers a fully frozen wall, $F\to\infty$ a fully catalytic one).

Viscous shear stress and heat flux in standard form:

$$\tau=\psi''+\cdots, \qquad q_1=T_1'+Le\,E_1c_1', \qquad q_2=T_2'+Le\,E_2c_2', \tag{2.26}$$

$$E_1=\frac{h_0}{c_pT_{20}(0)[1+c_{20}(0)]}, \qquad E_2=\frac{Ch_0}{Bc_p}. \tag{16}$$

Downstream self-similar limit ($x\to\infty$)

New variables, valid once $c_1(x,0)\to 0$ as $x\to\infty$:

$$n=\frac{\eta}{(3x)^{1/3}}, \qquad \psi=(3x)^{2/3}(\phi+J_2+\gamma_1 n), \qquad d=(3x)^{1/3}\gamma_1. \tag{2.27}$$

Resulting self-similar boundary value problem:

$$\begin{aligned}&\frac{\mathrm{d}^3\phi}{\mathrm{d}n^3}+2(\phi+\gamma_1 n+J_2)\frac{\mathrm{d}^2\phi}{\mathrm{d}n^2}-\left(\frac{\mathrm{d}\phi}{\mathrm{d}n}+2J\right)\frac{\mathrm{d}\phi}{\mathrm{d}n}+2\gamma_1\frac{\mathrm{d}\phi}{\mathrm{d}n}+2T_1(1+c_1)\phi\\ &\quad=(\gamma_1+J)^2-\frac{\mathrm{d}}{\mathrm{d}n}[T_1(1+c_1)]-2T_1(1+c_1)(c_1n+J_2),\end{aligned} \tag{2.28a}$$

$$\frac{1}{Sc}\frac{\mathrm{d}^2c_1}{\mathrm{d}n^2}+2(\phi+\gamma_1 n+J_2)\frac{\mathrm{d}c_1}{\mathrm{d}n}=0, \qquad \frac{1}{Pr}\frac{\mathrm{d}^2T_1}{\mathrm{d}n^2}+2(\phi+\gamma_1 n+J_2)\frac{\mathrm{d}T_1}{\mathrm{d}n}=0, \tag{17}$$

$$\phi=0, \quad \frac{\mathrm{d}\phi}{\mathrm{d}n}=-\gamma_1, \quad c_1=0, \quad T_1=\frac{1+\alpha}{[1+c_{20}(0)]} \quad (n=0), \tag{18}$$

$$\frac{\mathrm{d}\phi}{\mathrm{d}n}=0,\quad \frac{\mathrm{d}^2\phi}{\mathrm{d}n^2}=0,\quad c_1=c_{20}(0),\quad T_1=\frac{1}{[1+c_{20}(0)]}\quad (n\to\infty), \tag{19}$$

$$J=\int_0^n T_1(1+c_1)\,\mathrm{d}n,\qquad J_2=\int_0^\infty T_1(1+c_1)\,\mathrm{d}n. \tag{2.28}$$

A distinctive feature of this problem is the *absence* of a longitudinal pressure-gradient term; it involves an additional unknown parameter (eigenvalue) γ_1, determined by the solution itself (see §6 below for its numerical values).[1]

Asymptotic (downstream) shear stress and heat flux:

$$\tau=\psi'=\frac{\mathrm{d}^2\phi}{\mathrm{d}n^2}+T_1(1+c_1),\qquad q_1=\frac{1}{(3x)^{1/3}}\left(\frac{\mathrm{d}T_1}{\mathrm{d}n}+Le\,E_1\frac{\mathrm{d}c_1}{\mathrm{d}n}\right). \tag{2.29}$$

This shows explicitly that, far downstream, the shear stress τ returns to its undisturbed upstream value, while the heat flux q_1 decays algebraically as $(3x)^{-1/3}$.

7 Section 3: Region 4 — the immediate neighbourhood of the jump point

Matching conditions to the near-wall part of region 3 as $x_3\to -0$ ($A_w=(\partial u_3/\partial y_3)_w$):

$$u_3\to A_w y_3,\quad v_3=0,\quad p_3=p_w,\quad \rho_3=\frac{p_0}{T_{20}(0)[1+c_{20}(0)]},\quad T_{30}\to T_{20}(0),\quad c_{30}\to c_{20}(0). \tag{3.1}$$

Region-4 coordinates and expansions (Δx its characteristic length, $\varepsilon^{3/2}<\Delta x<\varepsilon^{3/4}$):

$$x=1+\Delta x\,x_4,\quad y=\varepsilon\Delta x^{1/3}y_4,\quad u=\Delta x^{1/3}u_4+\cdots,\quad v=\frac{\varepsilon}{\Delta x^{1/3}}v_4+\cdots, \tag{3.2}$$

$$p=p_0+\varepsilon^{1/2}p_w+\Delta x^{2/3}p_4+\cdots,\quad \rho=\rho_4+\cdots,\quad T=T_{40}+\cdots,\quad c=c_{40}+\cdots. \tag{20}$$

The governing equations in region 4 are of the same form as (2.8), with the subscript 3 replaced by 4; the matching/initial conditions are of the same form as (2.9), (2.11), with 3 replaced by 4, $D\equiv 0$, and A replaced by A_w.

Compensation-regime interaction condition (the induced pressure disturbance vanishes to leading order — the region-4 disturbance's displacement is exactly cancelled by the adjacent part of region 3, so there is no interaction with the outer supersonic flow at this order, and the problem remains parabolic rather than becoming an elliptic interaction problem):

$$\rho_{20}(0)A_w v_4+\frac{\mathrm{d}p_4}{\mathrm{d}x_4}\to 0,\qquad y_4\to\infty. \tag{3.3}$$

Shear stress and heat flux in region 4:

$$\tau=\varepsilon\mu\frac{\partial u_4}{\partial y_4}+\cdots,\qquad -q=\frac{\varepsilon\mu}{\Delta x^{1/3}Pr}\left[c_p\frac{\partial T_{40}}{\partial y_4}+Le\,\frac{\partial c_{40}}{\partial y_4}\left(\int_0^{T_{40}}(c_{p1}-c_{p2})\,\mathrm{d}T+h_0\right)\right]. \tag{3.4}$$

[1] Equations (2.28a)–(2.28) are reproduced here as transcribed from the OCR text of the source, and that transcription contains three errors that matter for computation. J_2 is a double integral with upper limit n (equivalently $\mathrm{d}J_2/\mathrm{d}n=J$, $J_2(0)=0$), not a constant evaluated at $n\to\infty$; the $\gamma_1\,\mathrm{d}\phi/\mathrm{d}n$ term enters the left-hand side of (2.28a) as $-2\gamma_1\,\mathrm{d}\phi/\mathrm{d}n$, not $+2\gamma_1\,\mathrm{d}\phi/\mathrm{d}n$; and the coefficient of n in the last forcing term is γ_1, not c_1. The corrected system actually solved is given in Part III, §§12–13.

New variables reducing region 4 to self-similar form:

$$x = \frac{x_4}{x^*}, \quad n = \frac{1}{\eta^*(3x)^{1/3}} \int_0^{y_4} \rho_4 \, \mathrm{d}y_4, \quad \psi_4 = \psi^*(3x)^{2/3}(\phi + J_2), \quad \frac{\partial \psi_4}{\partial y_4} = \rho_4 u_4, \quad \frac{\partial \psi_4}{\partial x_4} = -\rho_4 v_4, \tag{3.5a}$$

$$p_4 = p^*(3x)^{2/3}\gamma_2, \qquad c_1 = c_{40}, \qquad T_1 = \frac{T_{40}}{T_{20}(0)[1 + c_{20}(0)]}, \tag{21}$$

$$\eta^* = (x^*)^{1/3}\rho_{20}(0)^{-1/3}A_w^{-1/3}, \quad p^* = (x^*)^{2/3}\rho_{20}(0)^{-1/3}A_w^{1/3}, \quad \psi^* = (x^*)^{2/3}\rho_{20}(0)^{-1/3}A_w^{4/3}. \tag{3.5}$$

Region-4 self-similar boundary value problem (governs the jump in flow functions across the discontinuity point):

$$\begin{aligned} &\frac{\mathrm{d}^3\phi}{\mathrm{d}n^3} + 2(\phi + J_2)\frac{\mathrm{d}^2\phi}{\mathrm{d}n^2} - \left(\frac{\mathrm{d}\phi}{\mathrm{d}n} + 2J\right)\frac{\mathrm{d}\phi}{\mathrm{d}n} + 2T_1(1 + c_1)\phi \\ &\qquad = J^2 - \frac{\mathrm{d}}{\mathrm{d}n}[T_1(1 + c_1)] - 2T_1(1 + c_1)(\gamma_2 + J_2), \end{aligned} \tag{3.6a}$$

$$\frac{1}{Sc}\frac{\mathrm{d}^2 c_1}{\mathrm{d}n^2} + 2(\phi + J_2)\frac{\mathrm{d}c_1}{\mathrm{d}n} = 0, \qquad \frac{1}{Pr}\frac{\mathrm{d}^2 T_1}{\mathrm{d}n^2} + 2(\phi + J_2)\frac{\mathrm{d}T_1}{\mathrm{d}n} = 0, \tag{22}$$

$$\phi = 0, \quad \frac{\mathrm{d}\phi}{\mathrm{d}n} = 0, \quad c_1 = 0, \quad T_1 = \frac{1 + \alpha}{[1 + c_{20}(0)]} \quad (n = 0), \tag{23}$$

$$\phi = \gamma_2, \quad \frac{\mathrm{d}^2\phi}{\mathrm{d}n^2} = 0, \quad c_1 = c_{20}(0), \quad T_1 = \frac{1}{[1 + c_{20}(0)]} \quad (n \to \infty), \tag{3.6}$$

where γ_2 is a further eigenvalue, again determined only by requiring the solution to be well-behaved, not prescribed in advance. This problem is solved once, at the jump, and its outcome supplies the starting values used to march the standard-form problem (2.25) forward into $x > 0$.

Shear stress and heat flux in region-4 standard form:

$$\tau = \frac{\mathrm{d}^2\phi}{\mathrm{d}n^2} + T_1(1 + c_1) + \cdots, \qquad -q = \frac{1}{(3x)^{1/3}}\left(\frac{\mathrm{d}c_1}{\mathrm{d}n} + Le\, E_1\frac{\mathrm{d}T_1}{\mathrm{d}n}\right) + \cdots. \tag{3.7}$$

8 Section 4: A yet-finer region, and why it is not needed here

The paper notes that a region smaller still than region 4 exists in principle — one with $\Delta x \sim \Delta y \sim O(\varepsilon^{3/2})$, where the complete (unsimplified) Navier–Stokes and mass-conservation equations apply and streamwise ("lengthwise") diffusion becomes important. In that innermost region the no-slip condition at the wall would also, strictly, break down (a finite jump in temperature or concentration induces a small slip velocity there). The paper argues, however, that accounting for this innermost region is only necessary for a *removable* discontinuity (a weak jump, small enough that its effect can be absorbed smoothly); for the genuine, finite jumps studied in this paper, region 4 as constructed above is the correct innermost scale, and no numbered equations are introduced for this even-finer region.

9 Section 5 (numerical method): brief note

The paper solves the standard-form system (2.25) numerically by marching in x, using the far-downstream self-similar solution (2.28) to check the $x \to \infty$ behaviour and the jump-point self-similar solution (3.6) to supply consistent starting values at $x = 0^+$; because the interaction condition (2.10) introduces a mild upstream influence (the problem is only weakly elliptic), an extra boundary condition ($p \to 0$ as $x \to \infty$) is imposed, and a two-region numerical procedure is used to handle the jump in wall conditions at $x = 0$. No further governing equations are introduced in the numerical-method discussion; consult the original paper's citations to Daniels (1974) and Lipatov & Neiland (1982) for the numerical technique itself.

10 Section 6: Numerical results

Calculations were carried out for three representative combinations of the governing similarity parameters F (the Damköhler number of eq. 2.25), α (the size of the wall temperature jump), and $c_{20}(0)$ (the upstream wall atom concentration):

Case 1. $F = 1/\sqrt{2}$, $\alpha = 0$, $c_{20}(0) = 1$. (A moderate increase in catalytic activity, no temperature change.)

Case 2. $F \to \infty$, $\alpha = 0$, $c_{20}(0) = 1$. (An ideally catalytic downstream surface, no temperature change.)

Case 3. $F = 0$, $\alpha = 1$, $c_{20}(0) = 1$. (A frozen, non-catalytic downstream surface, with the wall temperature doubled.)

Physical character of Cases 1–2 vs. Case 3. At constant wall temperature (Cases 1–2), the jump in catalytic activity depletes the atom concentration near the wall, which increases the local gas density and pulls the streamlines toward the plate — an effective shallow depression, producing a locally low-pressure (expansion-type) disturbance. Conversely, doubling the wall temperature at fixed (frozen, non-catalytic) composition (Case 3) decreases the local gas density and pushes the streamlines away from the plate — an effective shallow bump, producing a locally high-pressure (compression-type) disturbance.

Eigenvalues. The two self-similar sub-problems' eigenvalues were determined numerically as:

	γ_2 (region 4, eq. 3.6, at $x = 0^+$)	γ_1 (region 2/3, eq. 2.28, as $x \to \infty$)
Case 2	0.22912	0.46094
Case 3	−0.46593	−0.81717

(Case 1, with finite F, does not reduce to either pure self-similar limit and was integrated as a genuinely non-self-similar solution of the full standard-form problem (2.25).)

Shear stress (skin friction) τ. Ahead of the jump ($x \to 0^-$), τ increases for the low-pressure/expansion cases (Cases 1–2) and decreases for the compression case (Case 3). At the jump itself, evaluating the region-4 self-similar solution (3.6)/(3.7) as $x \to 0^+$ gives

$$\tau(0^+) = 0.720 \text{ (Case 2)}, \qquad \tau(0^+) = 0.964 \text{ (Case 3)},$$

in good agreement with direct numerical integration of the full standard-form problem (2.25). Far downstream, $\tau \to 1$ in all cases, recovering the undisturbed upstream shear stress. Case 1's shear-stress distribution is continuous through the jump (no discrete drop), whereas Cases 2 and 3 show a

sharp change at $x = 0$ consistent with the values above. Quantitatively: the largest possible sudden increase in catalytic activity considered (Case 2) raises the shear stress by a factor of about 1.5 just ahead of the jump ($x \to 0^-$); a doubling of the wall temperature (Case 3) lowers the shear stress ahead of the jump to about 0.4 of its undisturbed value; and the paper notes that a temperature increase by a factor of about 3.87 (well beyond the $\alpha = 1$, i.e. doubling, case actually computed here) would be sufficient to trigger the onset of boundary-layer separation.

Temperature and concentration gradients. The normal temperature gradient at the wall (T_1' ahead of the jump, denoted T_2' downstream in the paper's figures) is only mildly affected by the catalytic-activity jump (Cases 1–2), but a doubling of the wall temperature (Case 3) reduces it by about 20% just ahead of the jump. Downstream of a strongly heated jump, the hot surface is cooled by the oncoming flow, and the normal temperature gradient changes sign (becomes negative) immediately after the jump. Far downstream, all temperature-gradient perturbations decay to their undisturbed values. The normal atom-concentration gradient behaves qualitatively like the shear stress and temperature-gradient distributions ahead of the jump; downstream, a jump to higher catalytic activity (Cases 1–2) increases the magnitude of the normal concentration gradient by a full order of magnitude, while the temperature-doubling case (Case 3) affects it only slightly. In all cases, the perturbations to both gradients decay to their undisturbed downstream values far from the jump.

Qualitative comparison with prior weak-discontinuity theories. The paper notes that its results are functionally similar in form to earlier studies that considered only small ("removable") jumps in surface catalytic properties without accounting for the induced pressure feedback, but that a direct quantitative comparison is not possible because those studies used different initial conditions and length scales.

Part III
Numerical Implementation

11 Overview

This part documents a MATLAB implementation of the downstream self-similar eigenvalue problem, eq. (2.28), which governs the solution's behaviour far downstream of the temperature/catalycity discontinuity (Part I, §4; Part II, §6). The implementation is provided as `downstream_eigenvalue_v2.m`, with a companion driver script `run_eigenvalue_v2.m` that reproduces the checks described below.

12 Corrections identified during implementation

Implemented exactly as printed, eq. (2.28) does not yield a well-posed eigenvalue problem. Three corrections were needed. Each was found by going back to the original page images of the source paper instead of trusting the OCR text extraction from which Part II's compendium was built, since OCR is unreliable for notation this dense in subscripts and fractions:

1. **The definition of J_2.** The source paper defines

$$J = \int_0^n T_1(1 + c_1)\,\mathrm{d}n, \qquad J_2 = \int_0^n \int_0^n T_1(1 + c_1)\,\mathrm{d}n\,\mathrm{d}n,$$

   i.e. J_2 is a genuine *double* integral with upper limit n (equivalently, an ordinary state variable satisfying $J_2' = J$, $J_2(0) = 0$) — not a constant obtained by evaluating a related integral

at $n \to \infty$. This makes γ_1 the only free eigenvalue parameter in the problem; J_2 is fully determined by integration once J (hence $\Theta = T_1(1 + c_1)$) is known.

2. **Sign of the $\gamma_1\varphi'$ term.** Rearranging the printed equation

$$\varphi''' + 2(\varphi + \gamma_1 n + J_2)\varphi'' - (\varphi' + 2\gamma_1 + 2J)\varphi' + 2\Theta\varphi = (\gamma_1 + J)^2 - \Theta' - 2\Theta(\gamma_1 n + J_2)$$

to isolate φ''' gives $+2\gamma_1\varphi'$ on the right-hand side (from $+(\varphi' + 2\gamma_1 + 2J)\varphi'$); an earlier implementation carried this term with the opposite sign.

3. **Coefficient in the last forcing term.** The final term on the right-hand side is $-2\Theta(\gamma_1 n + J_2)$ — the coefficient multiplying n inside the parentheses is γ_1, not c_1; an earlier implementation used $c_1 n$ in this position.

13 Governing system as implemented

With these corrections, the implemented system uses the state vector $y = [\varphi, \varphi', \varphi'', c_1, c_1', T_1, T_1', J, J_2]$ (nine components) and the single unknown parameter γ_1:

$$\begin{gathered}\varphi''' = -2(\varphi + \gamma_1 n + J_2)\varphi'' + (\varphi' + 2\gamma_1 + 2J)\varphi' - 2\Theta\varphi + (\gamma_1 + J)^2 - \Theta' - 2\Theta(\gamma_1 n + J_2),\\ c_1'' = -2\,Sc\,(\varphi + \gamma_1 n + J_2)\,c_1', \qquad T_1'' = -2\,Pr\,(\varphi + \gamma_1 n + J_2)\,T_1',\\ J' = \Theta, \qquad J_2' = J,\end{gathered}$$

with boundary conditions as given in Part II, eq. (2.28): at $n = 0$, $\varphi = 0$, $\varphi' = -\gamma_1$, $c_1 = 0$, $T_1 = (1 + \alpha)/[1 + c_{20}(0)]$, $J = 0$, $J_2 = 0$; as $n \to \infty$ (a large truncation length L in practice), $\varphi' = 0$, $\varphi'' = 0$, $c_1 = c_{20}(0)$, $T_1 = 1/[1 + c_{20}(0)]$. This is ten conditions for nine state variables plus one parameter, exactly determined. The MATLAB code solves this with `bvp4c`. Its parameterized-BVP calling convention (`bvpinit` with a parameter argument) was chosen because it treats γ_1 as an unknown solved for together with the profiles, which avoids wrapping an outer shooting loop around the eigenvalue and the convergence fragility such loops bring.

14 Validation status

With the corrections above, the computed eigenvalue γ_1 is stable to six decimal places across truncation lengths from $L = 3$ to $L = 20$ (holding $Pr = Sc = 1$). That is the expected signature of a correctly posed eigenvalue problem, and a qualitative change from the divergent or drifting behaviour obtained before these corrections were made.

Exact quantitative agreement with the two published values of γ_1 (Part II, §6: 0.46094 and -0.81717) has not been confirmed. The obstacle is that the source paper never states its Prandtl number Pr or Schmidt number Sc. A search over (Pr, Sc) can bring the computed γ_1 close to either published value individually; for example, $(Pr, Sc) = (0.25, 8.0)$ gives $\gamma_1 = -0.816$ against the published -0.81717. The catch is that the best-fit (Pr, Sc) pairs for the two cases are inconsistent with each other by roughly a factor of 5 in Pr and 18 in Sc, which is not physically sensible: both cases describe the same gas mixture and must share its transport properties. The more plausible reading is that the individual close matches are coincidences, and they should not be quoted as validation. Final quantitative validation requires the original computation's (Pr, Sc) values, which are not recoverable from the text alone; the region-4 problem (3.6), which would provide an independent check through γ_2, has not yet been implemented.